Microeconomics

Lecture 3 · Duality and the Slutsky equation

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What does a price change actually do?

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- Separate substitution from purchasing power: the Slutsky equation.
- Test the model: restrictions on demand derivatives.

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Today: fix utility instead of money.

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The target is ordinal: a bundle's rank matters, not the numerical utility scale.

The expenditure minimisation problem

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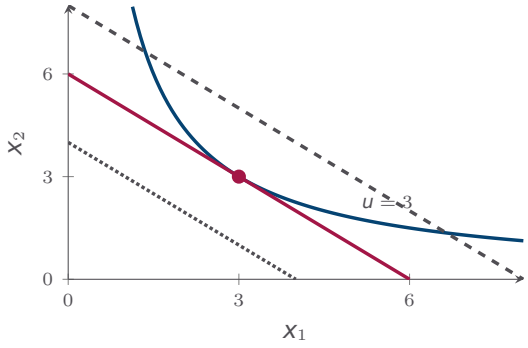
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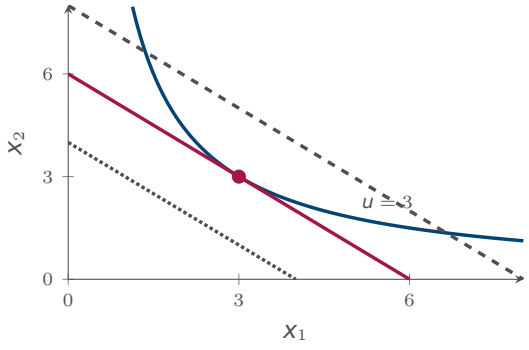
e: expenditure function. H: Hicksian demand correspondence; h if unique.

Move the budget line towards the origin



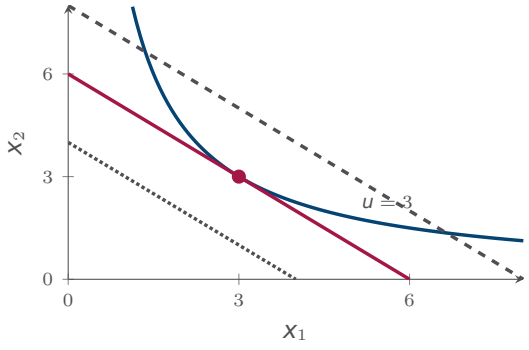
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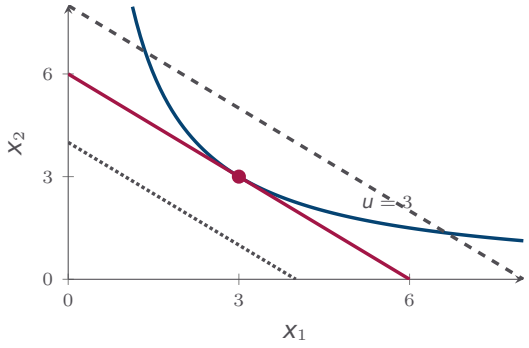
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Cost 4 cannot attain it.

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- Add locally nonsatiated preferences for the duality results.
- Add uniqueness and smoothness only where derivatives require them.

Does a minimum exist?

Proposition. If u is continuous, $p \gg 0$ and $\bar{u}$ is attainable, then $H(p, \bar{u}) \neq \varnothing$.

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How can we still use a compactness argument?

Find *one* feasible bundle and discard every bundle that costs more.

Existence: cut away expensive bundles

Choose $\widehat{x} \geq 0$ with $u(\widehat{x}) \geq \bar{u}$. Set

$$K = \{x \geq 0 : u(x) \geq \bar{u},\ p \cdot x \leq p \cdot \widehat{x}\}.$$

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Every excluded feasible bundle costs more than $p \cdot \widehat{x}$, so this is a global minimum.

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Otherwise, continuity gives $t < 1$ close to 1 with $u(th) > \bar{u}$.

But $p \cdot (th) < p \cdot h$: a cheaper feasible bundle. Contradiction.

Change the currency, not the chosen bundle

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Hicksian demand has degree 0 in prices; expenditure has degree 1.

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Two distinct minimisers have utility exactly $\bar{u}$. Their mixture would give more than $\bar{u}$, violating no excess utility.

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Otherwise a minimiser for $\bar{u}'$ would also minimise cost for $\bar{u}$, contradicting no excess utility.

Expenditure is concave in prices

Proposition. For a fixed attainable target and $0 \le t \le 1$,

$$e(tp + (1-t)q, \bar{u}) \ge t\,e(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

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The same bundle must be purchased at the mixed prices. Reoptimising separately can only be cheaper.

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Therefore

$$\begin{aligned} e(tp + (1-t)q, \bar{u}) &= (tp + (1-t)q) \cdot z \\ &\geq t\, e(p, \bar{u}) + (1-t) e(q, \bar{u}). \end{aligned}$$

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With continuous LNS utility, e is continuous in positive prices and interior attainable targets.

Buy a Cobb–Douglas standard of living

NES

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At $p = (4, 1)$ and $\bar{u} = 3$, what are the quantities and minimum cost?

Cobb–Douglas: the dual solution

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At $p = (4, 1)$ and $\bar{u} = 3$: $h = (3/2, 6)$, $e = 12$.

Compare $x_i(p, m) = m/(2p_i)$: Hicksian demand fixes $\bar{u}$, not m.

The kink is easy: perfect complements

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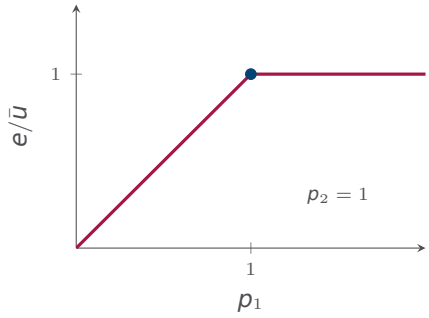
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Can the direct utility have a kink while expenditure is smooth in prices?

Yes. Here expenditure is linear in prices and compensated quantities do not change.

The cheapest route can switch

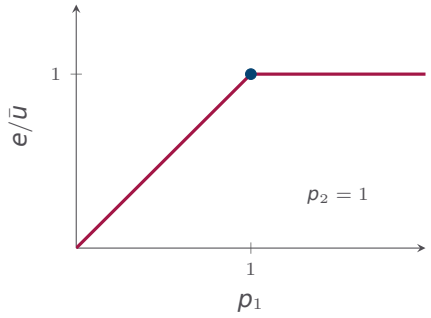
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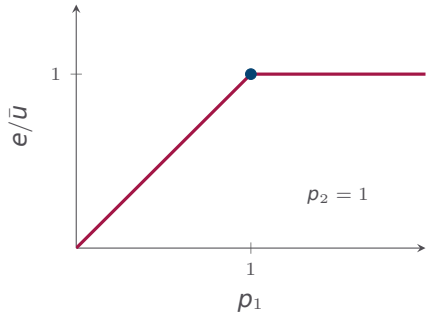


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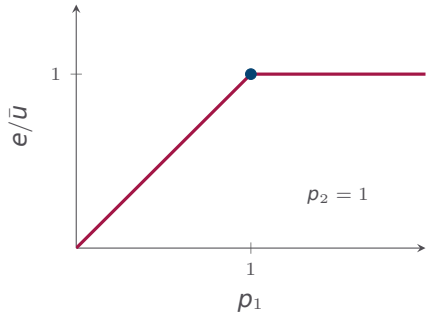


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$p_1 < p_2$: $h = (\bar{u}, 0)$.
$p_1 > p_2$: $h = (0, \bar{u})$.
$p_1 = p_2$: every $x \geq 0$ with $x_1 + x_2 = \bar{u}$ minimises cost.



Two problems, the same boundary

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The equalities hold for *sets of choices*, including nonunique optima.

Proof: the cheapest way to buy the maximum

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If $e(p, \bar{u}) < m$, a minimiser h attains at least $\bar{u}$ at cost below m.

Then h is also utility-maximising with a slack budget, contradicting LNS and Walras. Thus $e(p, v(p, m)) = m$.

Proof: the most utility the minimum cost buys

Choose $h \in H(p, \bar{u})$ and set $m = e(p, \bar{u})$. No excess utility gives $u(h) = \bar{u}$.

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That violates no excess utility. Hence $v(p, e(p, \bar{u})) = \bar{u}$.

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Substitute $m = e(p, \bar{u})$ for the other equality.

Check the identities by hand

For $u = \sqrt{x_1 x_2}$:

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Same choices at matching parameters; different responses when prices change.

Checkpoint

- Every price doubles; the utility target stays fixed. What happens to Hicksian quantities and expenditure?

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- If $\widetilde{u} = e^{u}$, which target describes the same standard of living as $\bar{u}$?
- If utility maximisation has two optimal bundles, can the corresponding expenditure problem have a unique solution?
- Can expenditure be concave in prices even when preferences are not convex?

Shephard’s lemma

Fix an attainable target and positive prices. If the Hicksian minimiser is unique, expenditure is differentiable in prices and

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = h_i(p, \bar{u}).$$

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Equivalently, wherever e is differentiable, every minimiser equals $\nabla_p e$.

Interpretation: a small price increase costs the compensated consumer approximately $h_i \, dp_i$.

Why the quantity appears in the derivative

Let $h = h(p, \bar{u})$ and choose $h_t \in H(p + te_i, \bar{u})$, where e_i is the ith unit vector.

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Divide by t and take limits; negative t gives the same derivative h_i.

What if demand is not unique?

For any $h \in H(p, \bar{u})$ and any $q \gg 0$,

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Use the demand correspondence at a kink; a single derivative cannot select the whole set.

Recover quantities from cost

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Substitutes: $e = \bar{u}\min\{p_1, p_2\}$; differentiate only within a strict price regime.

The compensated law of demand

Fix $\bar{u}$. Let $h \in H(p, \bar{u})$ and $k \in H(q, \bar{u})$. Then

$$(q - p) \cdot (k - h) \leq 0.$$

Why is it essential to hold the target fixed?

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No differentiability, uniqueness or convex preferences are required.

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For a differentiable unique demand: $\partial h_i / \partial p_i \leq 0$.

The matrix behind compensated substitution

Assume $e(\cdot, \bar{u})$ is C^2 locally. Define

$$S_{ij} = \frac{\partial h_i}{\partial p_j}, \qquad S = D_p h = D_{pp}^2 e.$$

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In particular, $S_{ii} \leq 0$. Zero effects are allowed.

Why symmetry and the quadratic inequality?

By Shephard, $h_i = e_{p_i}$. Mixed partials of a C^2 function commute:

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Concavity of e implies concavity of g, hence

$$g''(0) = z^\top D^2_{pp} e(p, \bar{u}) z \leq 0.$$

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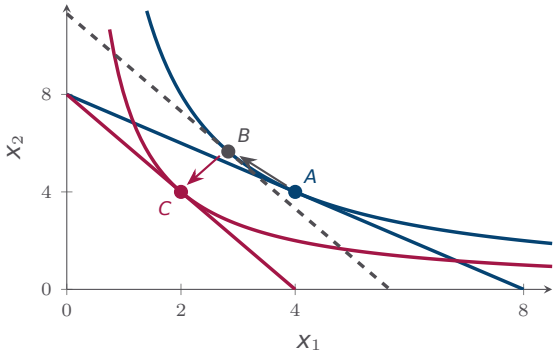
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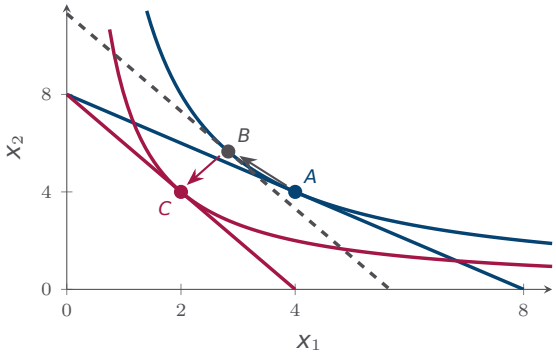
Relative prices drive compensated substitution; a change of currency does not.

A price increase: three choices on one picture



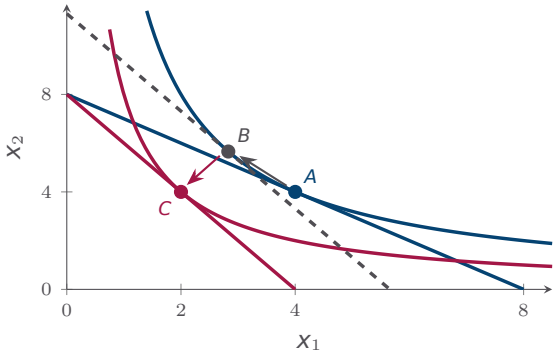
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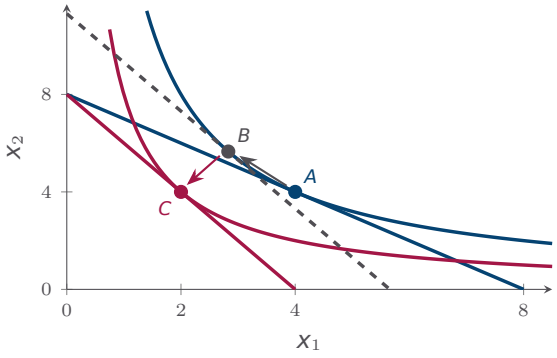
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Keep $u^0 = 4$:
$B = (2\sqrt{2}, 4\sqrt{2})$.
$A \to B$: substitution.
$B \to C$: income effect.

The choices match. Their price derivatives do not.

$$h(p, \bar{u}) = x(p, e(p, \bar{u})).$$

Which two terms appear when we differentiate the right-hand side?

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On the right, change prices *and adjust income* to $e(p, \bar{u})$.

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Slutsky: the chain rule does the work

Assume unique demands with the derivatives below. Start from

$$h_i(p, \bar{u}) = x_i(p, e(p, \bar{u})).$$

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Shephard gives $e_{p_j} = h_j$. At $\bar{u} = v(p, m)$, $h_j = x_j$ and $e = m$.

The Slutsky equation

At $\bar{u} = v(p, m)$:

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total price effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{purchasing-power term}}.$$

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The quantity multiplying the income derivative is x_j, not x_i.

When must ordinary demand slope down?

For the own price:

$$\frac{\partial x_i}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\leq 0} - x_i \frac{\partial x_i}{\partial m}.$$

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Giffen good: $\partial x_i / \partial p_i > 0$; the positive income effect dominates.

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At $p = (1,1)$, $m = 8$: total -4 = substitution -2 + income -2.

No total effect does not mean no substitution

Cobb–Douglas gives $x_1 = m/(2p_1)$, so $\partial x_1/\partial p_2 = 0$.

Does good 2 have no effect on the demand for good 1?

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The positive compensated substitution effect and negative income effect cancel exactly.

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Cobb–Douglas gives $x_1 = m/(2p_1)$, so $\partial x_1/\partial p_2 = 0$.

Does good 2 have no effect on the demand for good 1?

$$\frac{\partial h_1}{\partial p_2} = \frac{m}{4p_1p_2}, \qquad -x_2\frac{\partial x_1}{\partial m} = -\frac{m}{4p_1p_2}.$$

The positive compensated substitution effect and negative income effect cancel exactly.

At $p = (1, 1)$, $m = 8$: $0 = 2 - 2$.

Perfect complements: all income, no substitution

For $u = \min\{x_1, x_2\}$:

$$x_1 = x_2 = \frac{m}{p_1 + p_2}, \qquad h_1 = h_2 = \bar{u}.$$

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Ordinary demands fall together even though compensated quantities stay fixed.

Quasilinearity: income effects depend on the regime

$u(a, q) = a + 4\log(1 + q)$, $a + rq \leq m$, $a, q \geq 0$.

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“No income effect” applies within the interior numeraire regime.

Can a higher price make you buy more?

NES

At fixed tastes and income, a Giffen response means $\partial x_i / \partial p_i > 0$.

Can we construct a Giffen good with continuous, monotone, convex preferences?

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Yes. A smooth utility function is not needed for a smooth local demand.

A subsistence kink generates Giffen demand

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$$u(x) = \min\{x_1 + x_2,\ 3x_1 + 2x_2 - 10\}.$$

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The utility is continuous, strongly increasing and concave.

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At the original point, $h_{1,p_1} = 0$ and Slutsky gives $3.125 = 0 - (2.5)(-1.25)$.

Checkpoint

- At one point, $x_1 = 2$, $x_{1,m} = -1$ and $h_{1,p_1} = -3$. Is good 1 Giffen?

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- Can raising only p_1 increase h_1 at a fixed utility target?
- With quasilinear utility, can a low-income consumer's demand for the nonlinear good respond to income?

Substitutes and complements: specify the experiment

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With three or more goods, negative net cross-effects are possible.

Construct compensated effects from ordinary demand

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Test: $S = S^\top$, $z^\top S z \leq 0$, $Sp = 0$, plus homogeneity and Walras.

Cobb–Douglas passes the matrix test

For $x_i = m/(2p_i)$,

$$S = \frac{m}{4}\begin{pmatrix} -1/p_1^2 & 1/(p_1p_2) \\ 1/(p_1p_2) & -1/p_2^2 \end{pmatrix}.$$

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At $p = (1,1)$, $m = 8$: $S = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$.

Could these choices come from our consumer model?

NES

Suppose estimated demand is

$$x_1(p, m) = \frac{mp_1}{p_1^2 + p_2^2}, \qquad x_2(p, m) = \frac{mp_2}{p_1^2 + p_2^2}.$$

At $p = (1, 1)$ and $m = 2$, calculate the Slutsky matrix. Does the model survive?

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It is positive, smooth, homogeneous of degree 0, and $p \cdot x = m$.

At $p = (1, 1)$ and $m = 2$, calculate the Slutsky matrix. Does the model survive?

Start with $x = (1, 1)$ and $D_m x = (1/2, 1/2)^\top$.

A symmetric matrix can still fail

At $p = (1, 1)$, $m = 2$,

$$D_p x = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}.$$

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S is symmetric and $Sp = 0$. But $S_{11} = 1/2 > 0$.

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With $z = (1, -1)^\top$, $z^\top S z = 2 > 0$.

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With $z = (1, -1)^\top$, $z^\top S z = 2 > 0$.

The demand violates compensated demand restrictions. It cannot be generated by the maintained consumer model.

Checkpoint

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- A compensated own-price effect must be strictly negative.
- Gross cross-price effects must be symmetric.
- A Giffen good with positive consumption must be inferior at that point.

One structure, several ways to use it

$$u \longrightarrow v,\ x \qquad \longleftrightarrow \qquad e,\ h.$$

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Roy: differentiate indirect utility to recover ordinary demand.

Shephard: differentiate expenditure to recover compensated demand.

Slutsky: connect the two responses and obtain restrictions we can test.

Read next

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Chapter 8, *Slutsky Equation*.

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Homework 1 covers Lectures 1–2 and is due **14 November 2026**.

Reserve: compensation and equivalent variation

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CV restores old utility at new prices; EV removes the equivalent income at old prices.

Reserve: how much compensation?

For $u = \sqrt{x_1 x_2}$, $p^0 = (1, 1)$, $p^1 = (2, 1)$ and $m = 8$:

$$u^0 = 4, \qquad u^1 = 2\sqrt{2}.$$

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These are exact finite-change measures; a derivative is only a local approximation.

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To first order they agree: $de = h(p^0, u^0) \cdot dp = x^0 \cdot dp$.

Reserve: can we work backwards from demand?

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This is the integrability problem. Converse theorems require additional domain and regularity conditions.

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$$e = p_3\big[1 + \bar{u} - \log p_3 + \log(p_1 + p_2)\big].$$

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Goods 1 and 2 are net complements:

$$\frac{\partial h_1}{\partial p_2} = -\frac{p_3}{(p_1 + p_2)^2} < 0.$$

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Thus $\limsup e(p^k, \bar{u}^k) \leq p \cdot z$; let $z \to h$ to get the upper bound $e(p, \bar{u})$.

Reserve: continuity of expenditure

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Those cost bounds make minimisers locally bounded. A convergent subsequence is target-feasible by continuity, giving $\liminf e(p^k, \bar{u}^k) \geq e(p, \bar{u})$.